

# SQL Roadmap for Freshers

## Module 1 : Understanding Database

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### 1.1 What databases exist on this server?

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Scenario: You have connected to a company's MySQL server for the first time.

Question: How do you see all available databases?

SQL Query

```
SHOW DATABASES;
```

Explanation

Lists every database available on the server.

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### 1.2 Which database am I currently using?

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Scenario

You forgot which database is active.

Question

How do you check the current database?

SQL Query

```
SELECT DATABASE();
```

Explanation

Returns the active database.

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### 1.3 What is inside this database?

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Scenario

You want to explore a newly assigned database.

Question

How do you list all tables?

SQL Query

```
SHOW TABLES;
```

Explanation

Displays every table.

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## 1.4 What information does this table contain?

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Scenario

Before writing queries you need to know its columns.

Question

How do you inspect a table?

SQL Query

```
DESC employees;
```

Explanation

Shows columns, datatype and keys.

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## 1.5 How was this table created?

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Scenario

You need to know indexes, constraints and engine.

Question

How do you view the CREATE statement?

## SQL Query

```
SHOW CREATE TABLE employees;
```

## Explanation

Displays the complete CREATE TABLE script.

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## 1.6 What is Metadata?

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### Scenario

Instead of reading actual data, you want information about the database itself.

### Question

Where is metadata stored?

## SQL Query

```
SELECT *  
FROM INFORMATION_SCHEMA.TABLES;
```

## Explanation

Metadata stores information about databases, tables, columns and constraints.

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## 1.7 Which tables exist using Metadata?

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### Scenario

You are writing an admin tool.

### Question

How do you retrieve table names programmatically?

## SQL Query

```
SELECT TABLE_NAME
FROM INFORMATION_SCHEMA.TABLES
WHERE TABLE_SCHEMA = DATABASE();
```

Explanation

Reads table information from metadata.

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## Module 2 : Exploring Database

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### 2.1 How many tables are present?

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Scenario

Your project has many tables.

Question

How many tables exist?

SQL Query

```
SELECT COUNT(*)
FROM INFORMATION_SCHEMA.TABLES
WHERE TABLE_SCHEMA = DATABASE();
```

Explanation

Counts tables.

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### 2.2 Which table has the most rows?

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Scenario

You want to identify the largest table.

Question

How do you estimate table sizes?

## SQL Query

```
SELECT TABLE_NAME,  
TABLE_ROWS  
FROM INFORMATION_SCHEMA.TABLES  
WHERE TABLE_SCHEMA = DATABASE();
```

## Explanation

Shows estimated row count.

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## 2.3 Which columns exist in Employees?

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### Scenario

You need column names for documentation.

### Question

How do you list columns?

## SQL Query

```
SELECT COLUMN_NAME  
FROM INFORMATION_SCHEMA.COLUMNS  
WHERE TABLE_NAME='employees';
```

## Explanation

Returns every column.

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## 2.4 Which datatype does Salary use?

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### Scenario

Before calculations you need datatype information.

### Question

How do you find column datatype?

## SQL Query

```
SELECT COLUMN_NAME,  
DATA_TYPE  
FROM INFORMATION_SCHEMA.COLUMNS  
WHERE TABLE_NAME='employees';
```

## Explanation

Shows datatype of every column.

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# Module 3 : Retrieving Data

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## 3.1 Show all employees

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### Scenario

You need every employee.

### Question

Retrieve all rows.

## SQL Query

```
SELECT *  
FROM employees;
```

## Explanation

Returns every record.

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## 3.2 Show only employee names

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### Scenario

The report only needs names.

### Question

Retrieve one column.

SQL Query

```
SELECT name  
FROM employees;
```

Explanation

Returns only one column.

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### 3.3 Show unique departments

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Scenario

Many employees belong to the same department.

Question

Show distinct departments.

SQL Query

```
SELECT DISTINCT department  
FROM employees;
```

Explanation

Removes duplicates.

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### 3.4 Show first five employees

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Scenario

You only want sample data.

Question

Retrieve five rows.

SQL Query

```
SELECT *  
FROM employees  
LIMIT 5;
```

Explanation

Limits returned rows.

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## 3.5 Show latest employee

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Scenario

Find the newest employee.

Question

Retrieve last inserted row.

SQL Query

```
SELECT *  
FROM employees  
ORDER BY employee_id DESC  
LIMIT 1;
```

Explanation

Sorts descending and returns one row.

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## 3.6 Rename a column

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Scenario

Management wants "Employee Name" instead of "name".

Question

How do you rename a column in output?

SQL Query

```
SELECT name AS Employee_Name  
FROM employees;
```

Explanation

Alias changes only the output heading.

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